

INDUSTRY PROBLEM-SOLVING TASK

Java AWT, Applets, Event Handling and Layout Managers

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INDUSTRY TASK 1: Employee Performance & Payroll Dashboard

Problem Statement: A software company wants to develop a simple Employee Performance and Payroll Management System using Java AWT. The application should allow the HR team to enter employee details, process salary information, and generate a structured payroll report.

Functional Requirements

- Develop an AWT-based graphical user interface.
- Accept Employee ID, Employee Name, Department, Designation, Basic Salary, Working Days, and Performance Rating.
- Use suitable AWT components such as Label, TextField, Choice, Checkbox/CheckboxGroup, Button, TextArea, and List.
- Calculate Gross Salary, Performance Bonus, and Net Salary using suitable business rules defined by the student.
- Display the complete employee payroll report in a List or TextArea when Generate Report is clicked.
- Use GridBagLayout or BorderLayout for arranging the interface.
- Implement button events using ActionListener.
- Provide Submit/Generate Report and Reset functionality.
- Validate important inputs such as empty fields and numeric salary values.

Industry Context: The application simulates an HR payroll module used to capture employee information, perform basic payroll processing, and present a report for HR verification.

Expected Deliverables

- Java source code.
- Working GUI application with event handling.
- Sample input and output.
- Screenshot of the final application.
- Brief explanation of components, layout manager, and event handling used.

CODE:

```
import java.awt.*;
import java.awt.event.*;

class EmployeePayroll extends Frame implements ActionListener
{
    Label l1,l2,l3,l4,l5,l6,l7;
    TextField t1,t2,t3,t4,t5;
    Choice c1;
    Checkbox r1,r2,r3,r4;
    CheckboxGroup cg;
    Button b1,b2;
    TextArea ta;
    List list;

    EmployeePayroll()
    {
        Frame f = new Frame("Employee Performance & Payroll Dashboard");

        l1 = new Label("Employee ID");
        l2 = new Label("Employee Name");
        l3 = new Label("Department");
        l4 = new Label("Designation");
        l5 = new Label("Basic Salary");
        l6 = new Label("Working Days");
        l7 = new Label("Performance Rating");

        t1 = new TextField();
        t2 = new TextField();
        t3 = new TextField();
        t4 = new TextField();
        t5 = new TextField();

        c1 = new Choice();
        c1.add("IT");
        c1.add("HR");
        c1.add("Finance");
        c1.add("Marketing");

        cg = new CheckboxGroup();

        r1 = new Checkbox("Excellent",cg,false);
        r2 = new Checkbox("Good",cg,false);
        r3 = new Checkbox("Average",cg,false);
        r4 = new Checkbox("Poor",cg,false);

        b1 = new Button("Generate Report");
        b2 = new Button("Reset");

        ta = new TextArea();
        list = new List();

        f.setLayout(new GridBagLayout());
        GridBagConstraints g = new GridBagConstraints();

        g.fill = GridBagConstraints.HORIZONTAL;
        g.insets = new Insets(5,5,5,5);

        g.gridx=0; g.gridy=0;
        f.add(l1,g);
```

```

g.gridx=1;
f.add(t1,g);

g.gridx=0; g.gridy=1;
f.add(l2,g);
g.gridx=1;
f.add(t2,g);

g.gridx=0; g.gridy=2;
f.add(l3,g);
g.gridx=1;
f.add(c1,g);

g.gridx=0; g.gridy=3;
f.add(l4,g);
g.gridx=1;
f.add(t3,g);

g.gridx=0; g.gridy=4;
f.add(l5,g);
g.gridx=1;
f.add(t4,g);

g.gridx=0; g.gridy=5;
f.add(l6,g);
g.gridx=1;
f.add(t5,g);

g.gridx=0; g.gridy=6;
f.add(l7,g);
g.gridx=1;

Panel p = new Panel();
p.add(r1);
p.add(r2);
p.add(r3);
p.add(r4);
f.add(p,g);

g.gridx=0; g.gridy=7;
f.add(b1,g);

g.gridx=1;
f.add(b2,g);

g.gridx=0; g.gridy=8;
g.gridwidth=2;
f.add(ta,g);

b1.addActionListener(this);
b2.addActionListener(this);

f.setSize(600,600);
f.setVisible(true);

f.addWindowListener(new WindowAdapter()
{
    public void windowClosing(WindowEvent e)
    {
        f.dispose();
    }
}

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    });
}

public void actionPerformed(ActionEvent e)
{
    if(e.getSource()==b1)
    {
        if(t1.getText().equals("") ||
           t2.getText().equals("") ||
           t3.getText().equals("") ||
           t4.getText().equals("") ||
           t5.getText().equals(""))
        {
            ta.setText("Please fill all fields.");
            return;
        }

        try
        {
            double basic = Double.parseDouble(t4.getText());
            int days = Integer.parseInt(t5.getText());

            if(basic <= 0 || days <= 0 || days > 31)
            {
                ta.setText("Enter valid salary and working days.");
                return;
            }

            String rating = cg.getSelectedCheckbox().getLabel();

            double hra = basic * 0.20;
            double da = basic * 0.10;
            double gross = basic + hra + da;

            double bonus = 0;

            if(rating.equals("Excellent"))
                bonus = basic * 0.20;
            else if(rating.equals("Good"))
                bonus = basic * 0.10;
            else if(rating.equals("Average"))
                bonus = basic * 0.05;
            else
                bonus = 0;

            double net = gross + bonus;

            ta.setText("");

            ta.append("EMPLOYEE PAYROLL REPORT\n");
            ta.append("-----\n");
            ta.append("Employee ID      : " + t1.getText() + "\n");
            ta.append("Employee Name   : " + t2.getText() + "\n");
            ta.append("Department      : " + c1.getSelectedItem() + "\n");
            ta.append("Designation     : " + t3.getText() + "\n");
            ta.append("Basic Salary    : " + basic + "\n");
            ta.append("Working Days    : " + days + "\n");
            ta.append("Performance     : " + rating + "\n");
            ta.append("-----\n");
            ta.append("HRA              : " + hra + "\n");
            ta.append("DA               : " + da + "\n");

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        ta.append("Gross Salary      : " + gross + "\n");
        ta.append("Performance Bonus : " + bonus + "\n");
        ta.append("Net Salary       : " + net + "\n");
        ta.append("-----\n");

        list.add(t1.getText() + " - " + t2.getText() + " - " + net);
    }
    catch(Exception ex)
    {
        ta.setText("Please enter valid numeric values.");
    }
}

if(e.getSource()==b2)
{
    t1.setText("");
    t2.setText("");
    t3.setText("");
    t4.setText("");
    t5.setText("");
    ta.setText("");
    cg.setSelectedCheckbox(null);
    c1.select(0);
}

}

public static void main(String args[])
{
    new EmployeePayroll();
}

}

```

OUTPUT:

Employee Performance & Payroll Dashboard

Employee ID: E101

Employee Name: Bala

Department: IT

Designation: Developer

Basic Salary: 50000

Working Days: 26

Performance Rating: ☒ Excellent ☐ Good ☐ Average ☐ Poor

Basic Salary : 50000.0
 Working Days : 26
 Performance : Excellent

 HRA : 10000.0
 DA : 5000.0
 Gross Salary : 65000.0
 Performance Bonus : 10000.0
 Net Salary : 75000.0

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